Creating Your First

Data Connection

in Palantir Foundry

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S3 Bucket

Relational DB

REST API

Agenda

01 **Introduction & Setup**

Folder creation, Foundry overview

02 **Data Connection Basics**

Agents vs Direct Connection, architecture

03 **S3 Connection**

Connect to AWS S3, configure sync, import data

04 **Relational Database**

PostgreSQL connection, SSL certificates

05 **REST API Connection**

API source, Code Repository, Python transforms

Introduction

What You Will Learn

- Connect to Palantir-provided data sources
- Set up the three most common connection types
- Configure egress policies for secure access
- Sync external data into Foundry datasets
- Manage data pipelines via an intuitive interface

Data Sources Covered

AWS S3

Cloud object storage for flat files

PostgreSQL

Relational database (JDBC protocol)

REST API

HTTP endpoints via Code Repository

💡 No need to provide your own datasources — use Palantir-provided training sources.

Agents vs Direct Connection

🔒 Agent Connection

- Downloadable program installed in your org network
- Required for private networks & on-premises systems
- Communicates via encrypted outbound requests
- Requires provisioning, configuration & maintenance
- Best for: internal databases, on-prem file stores

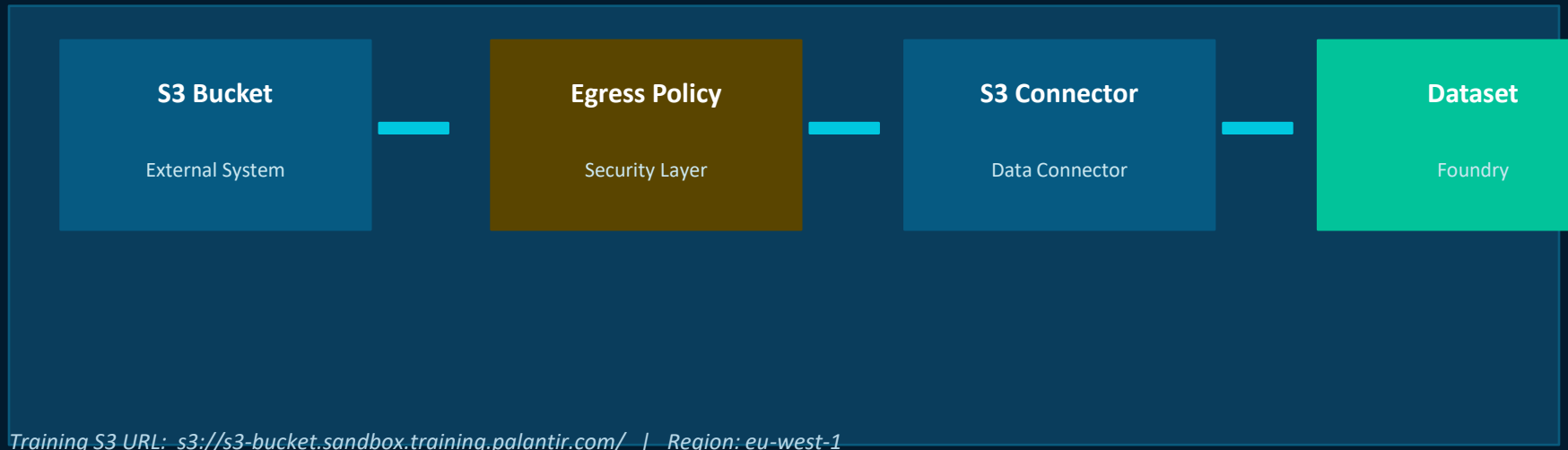
VS

🌐 Direct Connection

- For data sources accessible over the Internet
- Supports REST APIs, SFTP, cloud storage accounts
- No agent to provision, configure, or manage
- Automatically managed — excellent uptime
- Best for: S3, PostgreSQL (cloud), REST APIs ✓

Data Connection:
AWS S3 Bucket

Connect Foundry to an AWS S3 bucket and sync flat files into Foundry datasets



Training S3 URL: s3://s3-bucket.sandbox.training.palantir.com/ | Region: eu-west-1

S3 Connection — Step by Step

1. **Open Data Connection App**

   Applications → Data Connection → New source

2. **Select S3 Source Type**

   Choose S3 connector → Click Continue

3. **Choose Direct Connection**

   S3 is publicly accessible over the Internet

4. **Name & Save Source**

   "Data Connection Training - S3 (your_name)" → select project

5. **Enter Connection Details**

   URL, region, access key ID, secret access key

6. **Configure Egress Policy**

   Import existing or create new policy to whitelist the source

7. **Create Sync & Build**

   Explore source → select file → Create sync → run build

S3 — Connection Credentials

Connection Settings

URL	`s3://s3-bucket.sandbox.training.palantir.com/`
Region	`eu-west-1`
Auth Method	`Access key and secret`
Access Key ID	`AKIAWNSQVQU4RPCWEESQ`
Secret Access Key	`tWpDRP2QcCUl...m98  (encrypted)`

Egress Policy

- Whitelists the data source for secure Foundry access
- One-time setup — can be reused by later users
- Foundry auto-suggests the policy based on URL
- Import existing OR create new policy
- Address: s3-bucket.sandbox.training
- .palantir.com.s3.eu-west-1.amazonaws.com
- Port: 443

✓ After sync completes, open dataset → Apply a schema → data is ready for use in Foundry pipelines.

Data Connection:
Relational Database

Connect Foundry to a PostgreSQL database using the JDBC protocol with SSL encryption

Training DB: database.sandbox.training.palantir.com | Port: 5432 | DB: postgres

PostgreSQL — Configuration Details

Connection Settings

Host Type	Hostname
Hostname	database.sandbox.training.palantir.com
Port	5432
Database	postgres
Username	postgres_user
Password	speakfriendandenterdatabase

Two Egress Policies Required

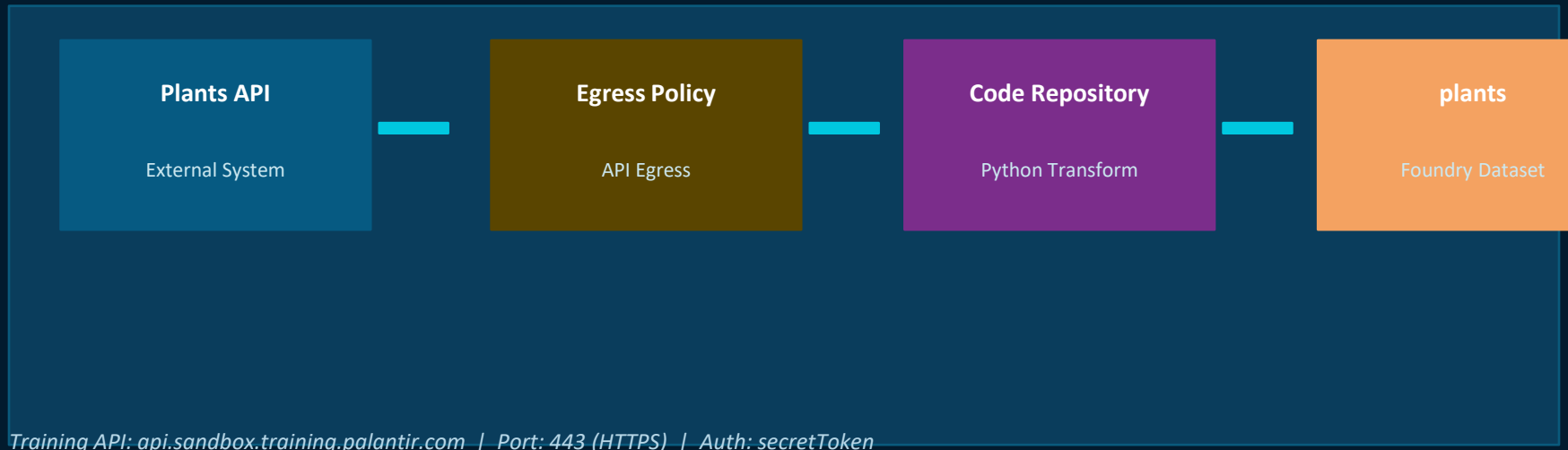
- Policy 1 — DNS: database.sandbox.training.palantir.com (Port 5432)
- Policy 2 — IP: 52.17.120.168 (Port 5432)

SSL Certificate Required

- PostgreSQL requires SSL for encrypted connection
- Download server-certificate.txt from lesson
- Alias: server.pem
- SSL Mode: require
- S3 does NOT need certificates (AWS handles it)

Data Connection:
REST API

Connect to HTTP endpoints using Foundry Code Repositories and Python transforms



Training API: api.sandbox.training.palantir.com | Port: 443 (HTTPS) | Auth: secretToken

REST API — Setup Workflow

1

Create REST API Source

Data Connection → New Source → REST API → Foundry worker

2

Configure Domain & Auth

URL: api.sandbox.training.palantir.com, Port 443, auth=None, secret=secretToken

3

Set Egress Policy

Import or create policy for api.sandbox.training.palantir.com:443

4

Enable Code Import

Allow source import into code repositories, set API name (camelCase)

5

Create Code Repository

Applications → Code Repositories → New Repo → Pipelines > Python → Spark Transform

6

Install Library

Libraries panel → search 'transforms-external-systems' → Add and install

7

Write Python Transform

Import source, call GET /secretPlants with auth header, write to output dataset

8

Commit & Build

Commit changes → Build → wait for successful run → view dataset

REST API — Python Transform Code

```python
from transforms.api import transform_df, Output
from transforms.external.systems import external_systems, Source

@external_systems(rest_api_source=Source("<<SOURCE RID>>"))
@transform_df(Output("<<OUTPUT_PATH>>"))
def compute(rest_api_source, ctx):
    url = rest_api_source.get_https_connection().url + "/secretPlants"
    auth_token = rest_api_source.get_secret("additionalSecretSecret")
    client = rest_api_source.get_https_connection().get_client()
    headers = { "Authorization" : auth_token }
    response = client.get(url, headers=headers)
    response.raise_for_status()
    json_content = response.json()
    return ctx.spark_session.createDataFrame(json_content)
```

Replace <<SOURCE RID>> with your source RID from the External Systems tab, and <<OUTPUT_PATH>> with your dataset path.

Key Concepts Summary

Data Connection

Foundry app that synchronizes data from external systems into Foundry datasets with a frontend interface.

Egress Policy

Network security policy that whitelists external data source addresses. Required for all Direct Connections.

Direct Connection

Used for Internet-accessible sources. No agent needed. Automatically managed by Foundry.

Agent Connection

Used for private networks & on-premises systems. Requires installing an agent within your org network.

Sync & Dataset Build

A Sync defines what data to bring in; a Build executes the import. Builds update dataset contents.

Code Repository (REST)

Python transforms in Foundry Code Repositories pull data from REST APIs using transforms-external-systems.

You've successfully learned how to:

- ✅ Set up data connections in Palantir Foundry
- ✅ Connect to AWS S3 cloud object storage
- ✅ Connect to PostgreSQL relational databases
- ✅ Connect to REST APIs via Code Repositories
- ✅ Configure egress policies for secure access

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